project.R

shwetasc03

2026-03-03

```
# Load required libraries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# Read dataset
data <- read.csv("/Users/shwetasc03/Downloads/star_digital.csv")

# Check structure
str(data)
```

```
## 'data.frame':    25303 obs. of  9 variables:
##  $ id      : int  545716 893524 1372718 971359 59999 842034 731724 49425 1351203 3
57681 ...
##  $ purchase: int  1 1 1 1 1 1 0 0 1 0 ...
##  $ test    : int  1 1 1 1 1 0 1 0 1 1 ...
##  $ imp_1   : int  0 1 0 14 0 0 2 0 97 0 ...
##  $ imp_2   : int  1 0 0 37 0 1 272 0 214 0 ...
##  $ imp_3   : int  0 0 0 1 0 0 0 0 0 0 ...
##  $ imp_4   : int  0 17 10 7 13 0 18 0 11 0 ...
##  $ imp_5   : int  0 0 0 0 0 0 0 0 3 0 ...
##  $ imp_6   : int  0 1 0 7 0 0 2 1 13 2 ...
```

```
summary(data)
```

```
##        id              purchase           test           imp_1
##  Min.   :     27   Min.   :0.0000   Min.   :0.000   Min.   :  0.0000
##  1st Qu.: 353880   1st Qu.:0.0000   1st Qu.:1.000   1st Qu.:  0.0000
##  Median : 708344   Median :1.0000   Median :1.000   Median :  0.0000
##  Mean   : 708953   Mean   :0.5029   Mean   :0.895   Mean   :  0.9309
##  3rd Qu.:1062738   3rd Qu.:1.0000   3rd Qu.:1.000   3rd Qu.:  0.0000
##  Max.   :1413367   Max.   :1.0000   Max.   :1.000   Max.   :296.0000
##      imp_2             imp_3                imp_4              imp_5
##  Min.   :  0.000   Min.   :  0.00000   Min.   :  0.000   Min.   : 0.00000
##  1st Qu.:  0.000   1st Qu.:  0.00000   1st Qu.:  0.000   1st Qu.: 0.00000
##  Median :  0.000   Median :  0.00000   Median :  0.000   Median : 0.00000
##  Mean   :  3.428   Mean   :  0.09477   Mean   :  1.589   Mean   : 0.04897
##  3rd Qu.:  2.000   3rd Qu.:  0.00000   3rd Qu.:  0.000   3rd Qu.: 0.00000
##  Max.   :373.000   Max.   :148.00000   Max.   :225.000   Max.   :51.00000
##      imp_6
##  Min.   :  0.000
##  1st Qu.:  0.000
##  Median :  1.000
##  Mean   :  1.783
##  3rd Qu.:  2.000
##  Max.   :404.000
```

```
#Question 1) Is online advertising effective? (Treatment vs Control)

#A Conversion Rate Comparison
data %>%
  group_by(test) %>%
  summarise(
    conversion_rate = mean(purchase),
    count = n()
  )
```

```
## # A tibble: 2 × 3
##    test conversion_rate count
##   <int>           <dbl> <int>
## 1     0           0.486  2656
## 2     1           0.505 22647
```

```
#B Difference in Means Test
t.test(purchase ~ test, data = data)
```

```
##
##  Welch Two Sample t-test
##
## data:  purchase by test
## t = -1.8713, df = 3309.2, p-value = 0.06139
## alternative hypothesis: true difference in means between group 0 and group 1 is no
t equal to 0
## 95 percent confidence interval:
##  -0.039289257  0.000916332
## sample estimates:
## mean in group 0 mean in group 1
##       0.4856928       0.5048792
```

```
#C Logistic Regression (Causal Effect)
model_q1 <- glm(purchase ~ test, data = data, family = binomial)
summary(model_q1)
```

```
##
## Call:
## glm(formula = purchase ~ test, family = binomial, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.05724    0.03882  -1.474   0.1404
## test         0.07676    0.04104   1.871   0.0614 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 35077  on 25302  degrees of freedom
## Residual deviance: 35073  on 25301  degrees of freedom
## AIC: 35077
##
## Number of Fisher Scoring iterations: 3
```

```
# Odds ratios
exp(coef(model_q1))
```

```
## (Intercept)        test
##   0.9443631   1.0797852
```

```
#Final answer>> Online advertising appears to have a positive effect on purchase beha
vior, increasing conversion rates by about 1.9 percentage points. However, the eviden
ce is only marginally significant, so the effectiveness of advertising is suggestive
rather than strongly conclusive.

#Question 2: Is there a frequency effect?

#A Create Total Impressions Variable
data$total_impressions <- rowSums(data[, c("imp_1","imp_2","imp_3","imp_4","imp_5","i
mp_6")])

#B Logistic Regression with Frequency
model_q2 <- glm(purchase ~ total_impressions,
                data = data,
                family = binomial)

summary(model_q2)
```

```
##
## Call:
## glm(formula = purchase ~ total_impressions, family = binomial,
##     data = data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -0.179392   0.014583  -12.30   <2e-16 ***
## total_impressions  0.029201   0.001294   22.56   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 35077  on 25302  degrees of freedom
## Residual deviance: 34212  on 25301  degrees of freedom
## AIC: 34216
##
## Number of Fisher Scoring iterations: 5
```

```
exp(coef(model_q2))
```

```
##       (Intercept) total_impressions
##         0.8357778         1.0296319
```

```
#C Optional: Diminishing Returns Test
data$total_impressions_sq <- data$total_impressions^2

model_q2_quad <- glm(purchase ~ total_impressions + total_impressions_sq,
                     data = data,
                     family = binomial)

summary(model_q2_quad)
```

```
##
## Call:
## glm(formula = purchase ~ total_impressions + total_impressions_sq,
##     family = binomial, data = data)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)          -2.406e-01  1.508e-02  -15.95   <2e-16 ***
## total_impressions     4.369e-02  1.646e-03   26.54   <2e-16 ***
## total_impressions_sq -1.158e-04  6.492e-06  -17.84   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 35077  on 25302  degrees of freedom
## Residual deviance: 33959  on 25300  degrees of freedom
## AIC: 33965
##
## Number of Fisher Scoring iterations: 4
```

```
#Question 3: Compare Sites 1–5 vs Site 6

#We compare effectiveness using impressions separately.
model_q3 <- glm(purchase ~ imp_1 + imp_2 + imp_3 + imp_4 + imp_5 + imp_6,
                data = data,
                family = binomial)
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
summary(model_q3)
```

```
##
## Call:
## glm(formula = purchase ~ imp_1 + imp_2 + imp_3 + imp_4 + imp_5 +
##     imp_6, family = binomial, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.200420   0.014696 -13.638  < 2e-16 ***
## imp_1       -0.006101   0.003738  -1.632   0.1026
## imp_2        0.014727   0.001760   8.366  < 2e-16 ***
## imp_3       -0.040928   0.016002  -2.558   0.0105 *
## imp_4        0.179939   0.006986  25.757  < 2e-16 ***
## imp_5       -0.449888   0.048144  -9.345  < 2e-16 ***
## imp_6        0.015993   0.003013   5.308 1.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 35077  on 25302  degrees of freedom
## Residual deviance: 33214  on 25296  degrees of freedom
## AIC: 33228
##
## Number of Fisher Scoring iterations: 6
```

```
exp(coef(model_q3))
```

```
## (Intercept)       imp_1       imp_2       imp_3       imp_4       imp_5
##   0.8183872   0.9939171   1.0148357   0.9598983   1.1971444   0.6376999
##       imp_6
##   1.0161214
```

```
#Compared to most of Sites 1–5, Site 6 performs better than Sites 1, 3, and 5. Howeve
r, Site 4 clearly outperforms Site 6 by a substantial margin.

#Question 4: Where should Star Digital allocate budget?

#Compare Combined Sites 1–5 vs Site 6
data$imp_1_5 <- rowSums(data[, c("imp_1","imp_2","imp_3","imp_4","imp_5")])

model_q4 <- glm(purchase ~ imp_1_5 + imp_6,
                data = data,
                family = binomial)

summary(model_q4)
```

```
##
## Call:
## glm(formula = purchase ~ imp_1_5 + imp_6, family = binomial,
##     data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.168177   0.014655 -11.476  < 2e-16 ***
## imp_1_5      0.032150   0.001464  21.958  < 2e-16 ***
## imp_6        0.014387   0.002929   4.911 9.05e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 35077  on 25302  degrees of freedom
## Residual deviance: 34186  on 25300  degrees of freedom
## AIC: 34192
##
## Number of Fisher Scoring iterations: 5
```

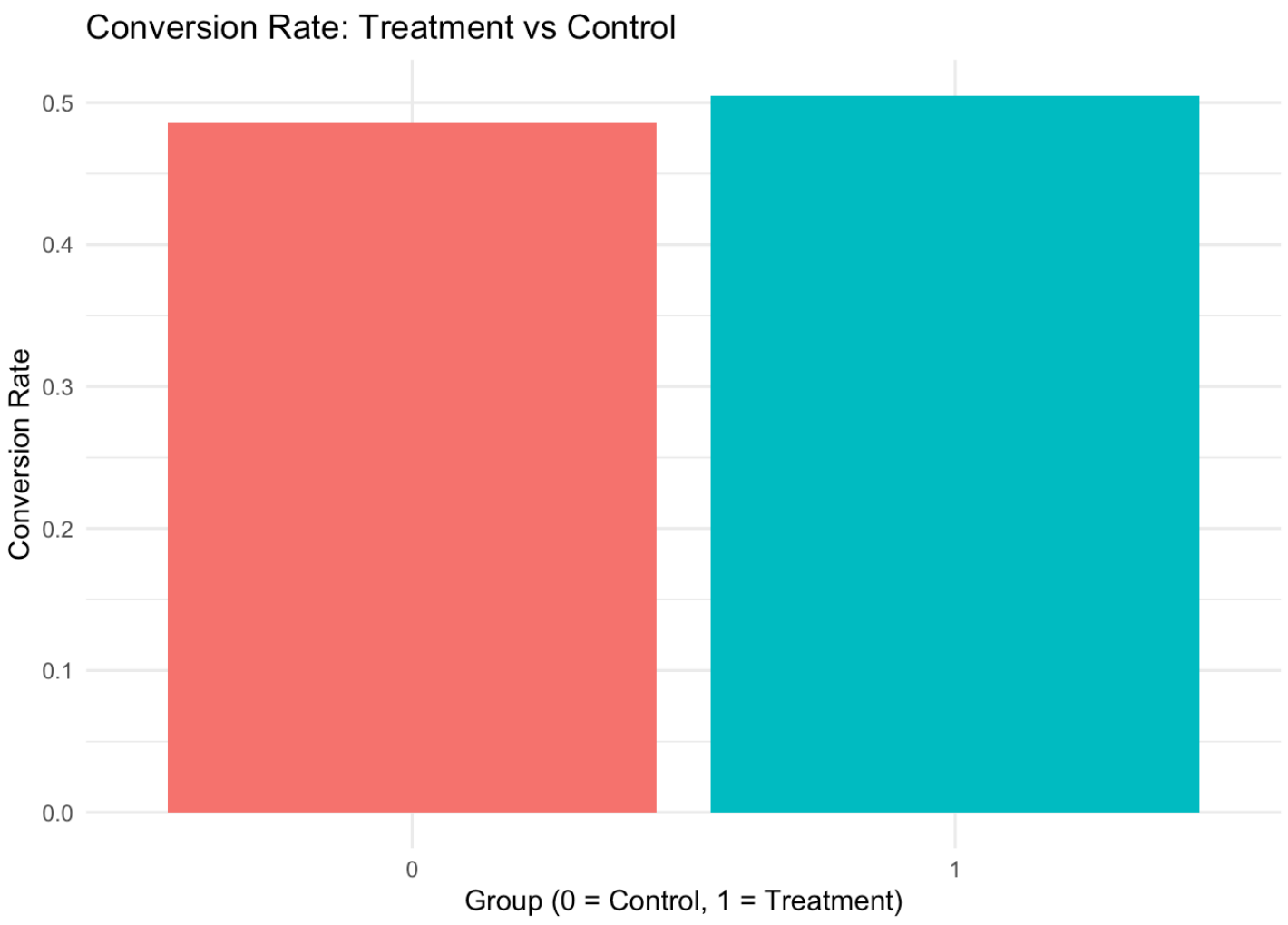
```
exp(coef(model_q4))
```

```
## (Intercept)     imp_1_5       imp_6
##   0.8452044   1.0326727   1.0144913
```

```
#Predicted Probability Comparison
data$predicted_prob <- predict(model_q4, type = "response")
mean(data$predicted_prob)
```

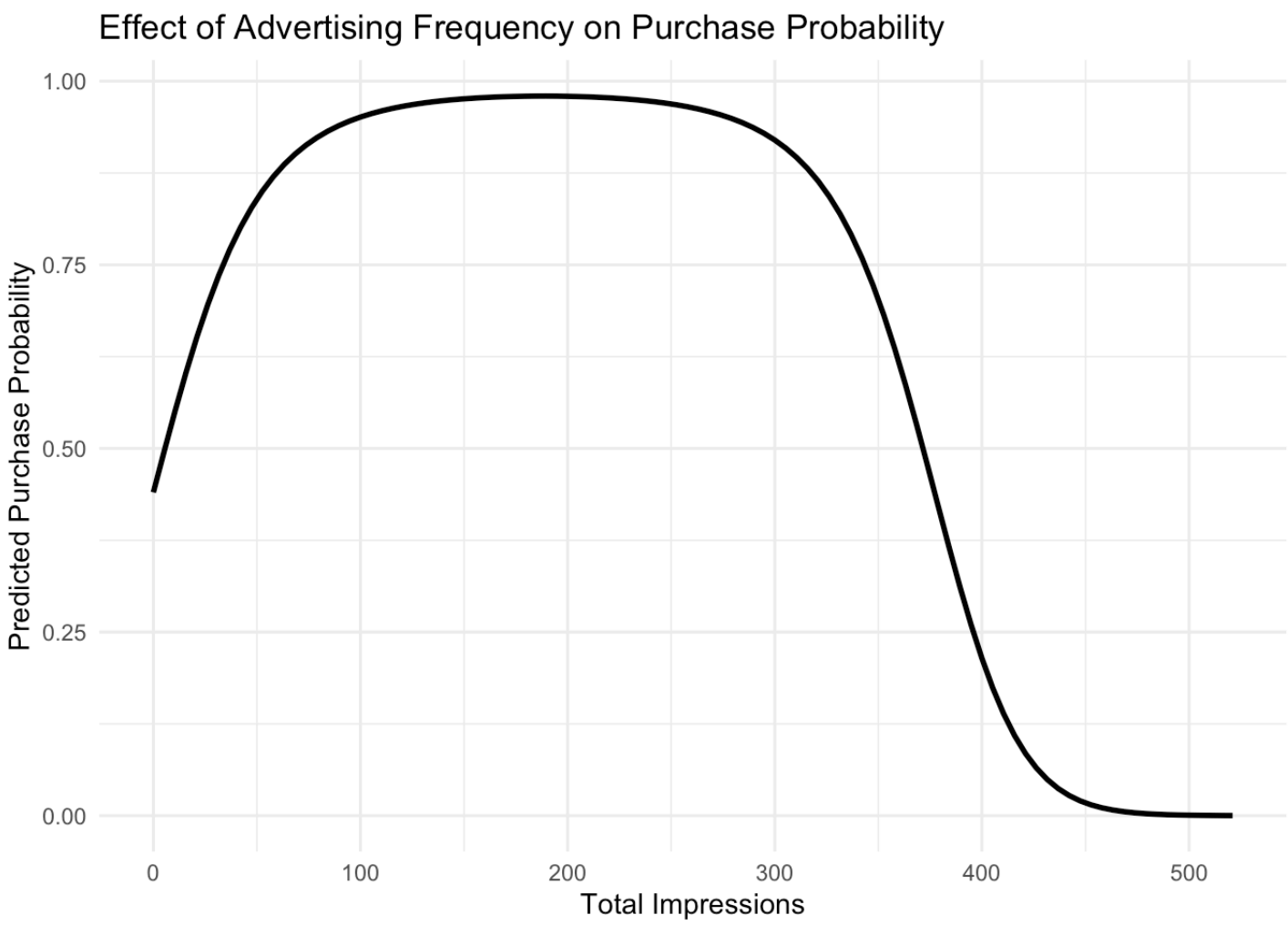
```
## [1] 0.5028653
```

```
#While both advertising channels are effective, Sites 1–5 collectively produce a larg
er marginal return per impression than Site 6. Therefore, Star Digital should allocat
e a greater share of its advertising budget to Sites 1–5, with emphasis on the highes
t performing site within that group.

library(ggplot2)

#Question 1 visual
# Conversion rate by treatment
conv_data <- data %>%
  group_by(test) %>%
  summarise(conversion_rate = mean(purchase))

ggplot(conv_data, aes(x = factor(test), y = conversion_rate, fill = factor(test))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Conversion Rate: Treatment vs Control",
    x = "Group (0 = Control, 1 = Treatment)",
    y = "Conversion Rate"
  ) +
  theme_minimal() +
  guides(fill = "none")
```

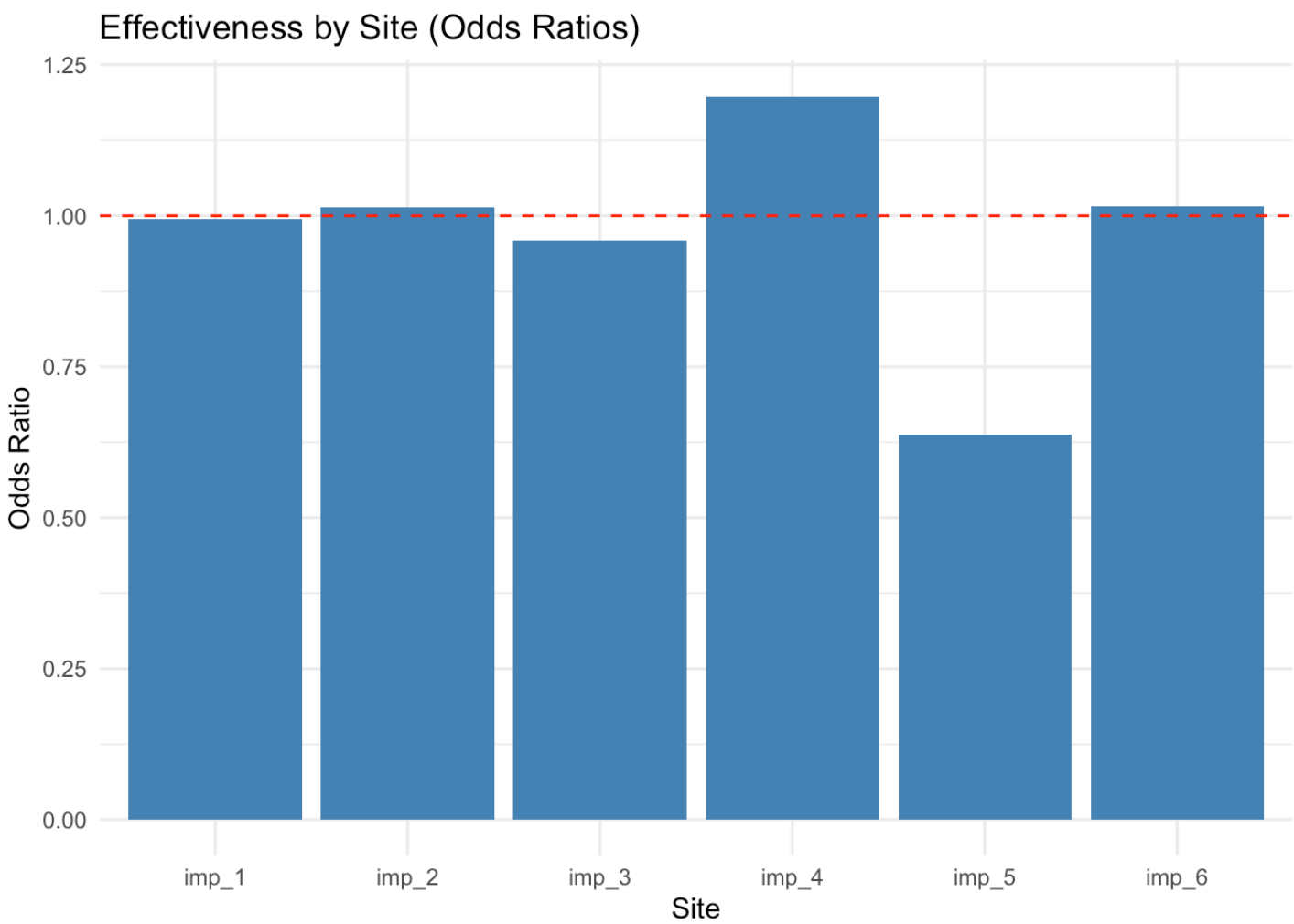

Conversion Rate: Treatment vs Control
Conversion Rate
0.5
0.4
0.3
0.2
0.1
0.0
0
1
Group (0 = Control, 1 = Treatment)

```
#Question 2 Visual
# Create prediction grid
new_data <- data.frame(
  total_impressions = seq(0, max(data$total_impressions), length.out = 100)
)

new_data$total_impressions_sq <- new_data$total_impressions^2

new_data$predicted_prob <- predict(model_q2_quad,
                                   newdata = new_data,
                                   type = "response")

ggplot(new_data, aes(x = total_impressions, y = predicted_prob)) +
  geom_line(size = 1) +
  labs(
    title = "Effect of Advertising Frequency on Purchase Probability",
    x = "Total Impressions",
    y = "Predicted Purchase Probability"
  ) +
  theme_minimal()
```

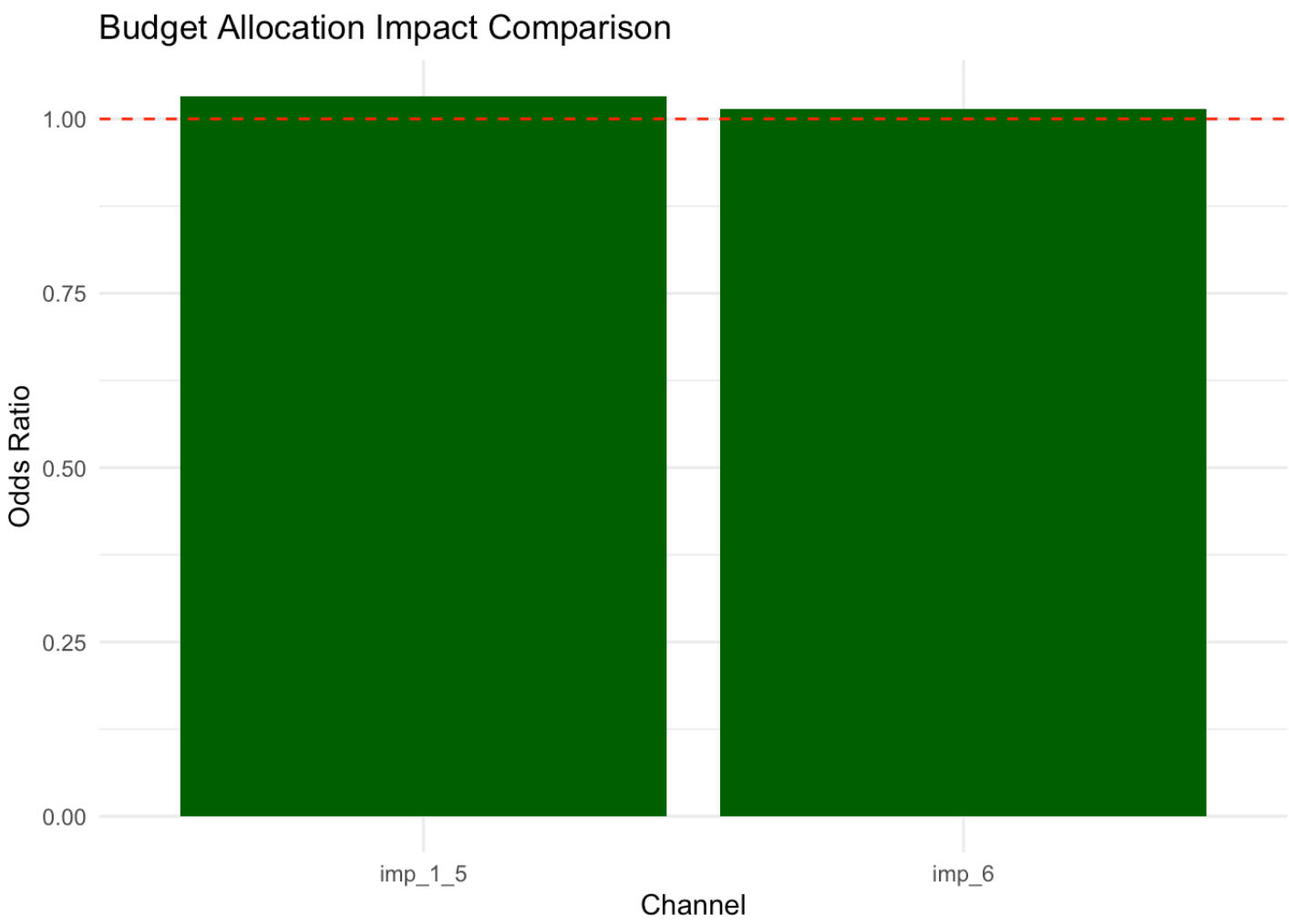
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
#Question 3 visual
# Extract odds ratios
odds_ratios <- exp(coef(model_q3))[-1]

site_data <- data.frame(
  site = names(odds_ratios),
  odds_ratio = odds_ratios
)

ggplot(site_data, aes(x = site, y = odds_ratio)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  labs(
    title = "Effectiveness by Site (Odds Ratios)",
    x = "Site",
    y = "Odds Ratio"
  ) +
  theme_minimal()
```




```
#Question 4 visual
budget_odds <- exp(coef(model_q4))[-1]

budget_data <- data.frame(
  channel = names(budget_odds),
  odds_ratio = budget_odds
)

ggplot(budget_data, aes(x = channel, y = odds_ratio)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  geom_hline(yintercept = 1, linetype = "dashed", color = "red") +
  labs(
    title = "Budget Allocation Impact Comparison",
    x = "Channel",
    y = "Odds Ratio"
  ) +
  theme_minimal()
```


Budget Allocation Impact Comparison
1.00
0.75
0.50
0.25
0.00
Odds Ratio
imp_1_5
imp_6
Channel